CREDIT REFERENCE · JULY 2026 EDITION

LEED v5 BD+C, but better looking

A reference for the LEED v5 AP BD+C exam. One prerequisite or credit per page, and two for the longest few, with its intent, points for New Construction and Core and Shell, every option and path with its requirement, the numbers worth memorizing, the standards it runs on and the credits it shares work with.

Integrative Process, Planning and Assessments	1 NC · 7 CS
Location and Transportation	15 NC · 15 CS
Sustainable Sites	11 NC · 11 CS
Water Efficiency	9 NC · 8 CS
Energy and Atmosphere	33 NC · 27 CS
Materials and Resources	18 NC · 21 CS
Indoor Environmental Quality	13 NC · 11 CS
Project Priorities	10 NC · 10 CS

Built from the same data as the LEED v5 Study app. Checked against the July 2026 rating system. 51 prerequisites and credits.

IP

Integrative Process, Planning and Assessments

1 point in New Construction · 7 in Core and Shell · the starting line

IPp1	Climate Resilience Assessment	Required
IPp2	Human Impact Assessment	Required
IPp3	Carbon Assessment	Required
IPp4	Tenant Guidelines	Required
IPc1	Integrative Design Process	1 / 1
IPc2	Green Leases	– / 6

Points shown as NC / CS. A dash means the credit is not offered in that system.

IPP1 · PREREQUISITE

Climate Resilience Assessment

REQUIRED | QUALITY OF LIFE

Assess observed and projected natural hazards so the design can reduce the project's vulnerability to them.

REQUIREMENTS

REQ

Climate and natural hazard assessment

Rate every relevant site hazard using an IPCC SSP scenario across the service life. Cover hazard level, exposure, sensitivity, adaptive capacity and overall risk.

AND

REQ

Two priority hazards

Carry at least two priority hazards into design strategies. These same hazards drive SSc4.

NUMBERS TO KNOW

2	Minimum priority hazards, which also drive the site credit

STANDARDS

IPCC shared socioeconomic pathways
Climate scenarios

TRANSFERS

→ SSc4
The two priority hazards from IPp1 drive SSc4.

Resilience line
IPp1 → SSc4 → EAc6 → EQc4

FROM THE PRACTICE TESTS

7 QUESTIONS

Along with the IPCC emissions scenario, what must the team specify for each priority hazard?
Shared socioeconomic pathways

Besides building function, what other impact must each priority hazard be assessed for?
Impact on the site during construction

Is a cost-benefit analysis of resilience strategies required for each hazard?
No

What is the difference between hazard level and hazard risk rating?
Level is intensity (low, medium, high). Risk rating is likelihood, from hazard mitigation plans.

Which time horizons of natural hazards must the assessment identify?
Observed, projected and future

Beyond listing and rating hazards, what does the prerequisite expect?
Use it to inform design and O&M, where possible

A school with a 100-year service life should pick which emissions scenario?
A higher-warming scenario NOT IN RS

IPP2 · PREREQUISITE

Human Impact Assessment

REQUIRED | QUALITY OF LIFE

Understand the social context of the community, workforce and supply chain before design decisions get made.

REQUIREMENTS

REQ

Human impact assessment

Draw on demographics, local infrastructure and land use, human use and health impacts, and occupant experience.

TRANSFERS

→ SSc5

International Tree Equity uses the Human Impact Assessment.

FROM THE PRACTICE TESTS

4 QUESTIONS

Which methods does LEED suggest for gathering human impact data?

Site analysis, outreach, census, GIS, partners NOT IN RS

Which four categories does the human impact assessment draw from?

Demographics, infrastructure, human use, occupant experience

What does the occupant experience category cover?

Daylight, views and operable windows, air and water conditions, and adjacent soundscapes, lighting and wind

What referenced standards apply?

None NOT IN RS

IPP3 · PREREQUISITE

Carbon Assessment

REQUIRED | DECARBONIZATION

See long-term emissions from operations, refrigerants and embodied carbon in a single projection.

REQUIREMENTS

REQ

25-year carbon projection

USGBC generates it from EAp1, EAp5 and MRp2 data. LTc4 transportation data is optional.

NUMBERS TO KNOW

25 yr	Carbon projection horizon

TRANSFERS

← EAp1
Operational energy data feeds the 25-year carbon projection.

← EAp5
Refrigerant data feeds the 25-year carbon projection.

← MRp2
Embodied carbon data feeds the 25-year carbon projection.

← LTc4
Transportation data is an optional input to the projection.

Carbon line
IPp3 → EAp1 → MRp2 → MRc2 → EAc1 → EAc3 → EAc4

FROM THE PRACTICE TESTS

1 QUESTIONS

Does the team run new calculations for the carbon assessment?
No, USGBC builds it

IPP4 · PREREQUISITE · CS ONLY

Tenant Guidelines

REQUIRED | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Hand the base building's sustainability features and goals to tenants so their fit-outs finish the job.

REQUIREMENTS

REQ

Tenant guidelines

Share features, goals, fit-out guidance and a point of contact with every tenant before lease signing.

TRANSFERS

→ IPc2

Green leases make binding what the tenant guidelines recommend.

FROM THE PRACTICE TESTS

4 QUESTIONS

Tenant guideline recommendations must align with what?
The base building's attempted credits

Who serves as the tenant guidelines point of contact?
Someone from the base building team or the owner

When must every tenant receive the tenant guidelines?
Before signing the lease

How do the IPp4 tenant guidelines differ from an IPc2 green lease?
Guidelines inform and recommend. Leases bind tenants contractually.

IPC1 · CREDIT

Integrative Design Process

NC 1 PTS | CS 1 PTS | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Find synergies across disciplines from predesign onward and set measurable goals for all three impact areas.

REQUIREMENTS

REQ
Integrated team
Convene an interdisciplinary team with diverse perspectives and organized, equitable facilitation.

AND

REQ
Design charrette
Owner plus at least four key perspectives, held in predesign or early design.

AND

REQ
LEED goal setting
Specific, measurable goals for decarbonization, quality of life and ecological restoration, written into the OPR.

NUMBERS TO KNOW

Owner + 4	Perspectives required at the design charrette

FROM THE PRACTICE TESTS

3 QUESTIONS

Which document must hold the specific, measurable LEED goals?
Owner's project requirements

Which phase of the integrative process is the most important?
Discovery NOT IN RS

What role does the DOE charrette handbook play?
A supporting resource, not a referenced standard NOT IN RS

IPC2 · CREDIT · CS ONLY

Green Leases

NC NONE | CS 6 PTS | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Use lease terms to make tenants complete what the base building started.

REQUIREMENTS

OPT 1 · 1–6 PTS

Standard Green Lease

Fit-out standards in every future lease. Best practices added set the points. Fewer than 3 = 1, 4–6 = 2, 7–9 = 3, 10–12 = 4, 13–15 = 5, 16+ = 6.

AND/OR

OPT 2 · 1 PTS

Executed Standard Green Lease

One extra point for an executed tenant green lease or tenant attestation that meets Option 1.

OR

OPT 3 · 3–6 PTS

Green Lease Leaders Recognition

Earn Green Lease Leaders recognition.

3 PTS
Silver

OR

4 PTS
Gold

OR

6 PTS
Platinum

TRANSFERS

← IPp4
Green leases make binding what the tenant guidelines recommend.

FROM THE PRACTICE TESTS

8 QUESTIONS

What does Option 1 reward, and what does Option 2 reward?
Standard lease / executed lease

How many prerequisites must the green lease hold tenants to, and is water metering (WEp1) one of them?
9, and no

What problem do green leases address when tenants pay the utility bills?
The split incentive NOT IN RS

Is MRp1 Planning for Zero Waste Operations among the tenant prerequisites? Is EQp3 No Smoking?
MRp1 yes, EQp3 no

If the tenant has not finalized the lease, what can it submit instead?
Signed, dated letter of attestation

Which is a green lease best practice, energy training for brokers or tenants buying RECs for fit-out?
Broker energy training

Can a New Construction project pursue IPc2 Green Leases?
No, it is Core and Shell only

What are the Green Lease Leaders recognition levels and points?
Silver 3, Gold 4, Platinum 6

LT

Location and Transportation

15 points in New Construction · 15 in Core and Shell · the line to Downtown

LTc1	Sensitive Land Protection	1 / 1
LTc2	Equitable Development	2 / 2
LTc3	Compact and Connected Development	6 / 6
LTc4	Transportation Demand Management	4 / 4
LTc5	Electric Vehicles	2 / 2

Points shown as NC / CS. A dash means the credit is not offered in that system.

LTC1 · CREDIT

Sensitive Land Protection

NC 1 PTS | CS 1 PTS | ECOLOGICAL CONSERVATION AND RESTORATION

Keep the development footprint off land that provides critical ecosystem services.

REQUIREMENTS

OPT 1 · 1 PTS

Previously Developed Sites

Locate the development footprint on previously developed land.

OR

OPT 2 · 1 PTS

Previously Undeveloped Sites

Avoid prime farmland, floodplains, notable habitat, 100 ft of water bodies and 50 ft of wetlands. Protect 40% of steep slopes, 60% above 25%.

NUMBERS TO KNOW

50 ft	Wetland buffer on undeveloped sites
100 ft	Water body buffer on undeveloped sites

STANDARDS

NatureServe, U.S. Endangered Species Act
Imperiled species and habitat

7 CFR 657.5, NRCS soil surveys
Prime farmland

FROM THE PRACTICE TESTS

6 QUESTIONS

With no legally adopted flood map, where must the development footprint sit?
Outside the 100-year floodplain

What share of unstable slopes between 15% and 25% must be protected?
40%

Which NatureServe ranks mark notable habitat, and does G3 count?
GH, G1 and G2. G3 does not.

What share of unstable slopes steeper than 25% must be protected?
60%

Which U.S. references verify sensitive land?
NRCS soil surveys, NatureServe, U.S. Endangered Species Act

Which farmland categories must be avoided?
Prime, unique, and statewide or local importance, per 7 CFR 657.5

LTC2 · CREDIT

Equitable Development

NC 2 PTS | CS 2 PTS | QUALITY OF LIFE

Support local economies and community vitality through where and how the project gets built.

REQUIREMENTS

OPT 1 · 1–2 PTS

Priority Sites

PATH 1 · 2 PTS

Brownfield Remediation

Build on a site the authority requires remediated, and remediate to its satisfaction.

OR

PATH 2 · 1 PTS

Historic Location

Locate in a historic district identified by the local government.

AND/OR

OPT 2 · 1–2 PTS

Housing and Jobs Proximity

PATH 1 · 1 PTS

Support Local Economy

Hire 15% of construction jobs from people living in the project's administrative district.

OR

PATH 2 · 2 PTS

Location-Efficient Affordable Housing

10% of residential area affordable. Rentals at 60% AMI for 15 years, for-sale at 80% AMI. Plus transit, walkability or a 1.2 jobs-to-housing ratio.

OR

OPT 3 · 2 PTS

Equitable Construction

15% of construction hours by registered apprentices, or one on-site life-skills course a month.

AND/OR

OPT 4 · 2 PTS · SCHOOLS

Equitable Access to Resources

PATH 1 · 1 PTS

Public Use Spaces

Open at least three school spaces to shared public use, with toilets after hours.

AND/OR

PATH 2 · 1 PTS

Community Partnership

Contract at least two dedicated-use spaces with community organizations.

AND/OR

OPT 5 · 1 PTS · DATA CTR, WAREHOUSE

Sensitive Project Location

Meet Option 1, 2 or 3 and sit at least 300 ft from homes, schools, parks and other sensitive receptors.

STANDARDS

DOL registered apprenticeships
Workforce training, 15% of construction hours

FROM THE PRACTICE TESTS

6 OF 9 QUESTIONS, THE REST ARE IN THE APP

How many space types for the public use spaces path, and for the community partnership path?
3 shared / 2 dedicated

How often must life-skills courses be scheduled?
One a month during construction

If the shared spaces sit in a separate school building, what conditions apply?
3 space types within 0.25 mi on campus

What jobs-to-housing ratio must be exceeded, and within what walking distance?
More than 1.2:1 within 0.5 mi

How far must a data center, warehouse or distribution center sit from sensitive receptors' property lines?
300 ft

How long must rental units stay affordable?
At least 15 years, with existing units excluded

LTC3 · CREDIT

Compact and Connected Development

NC 6 PTS | CS 6 PTS | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Build where density, infrastructure and transit already exist, cutting vehicle distance and land use.

REQUIREMENTS

OPT 1 · 1–2 PTS

Surrounding Density

Within 0.25 mi. Combined 22,000 / 35,000 sf per acre, or 7 / 12 DU per acre and FAR 0.5 / 0.8.

AND/OR

OPT 2 · 1–4 PTS

Access to Transit

PATH 1 · 1–4 PTS

Public Transit Service

Bus within 0.25 mi, or rail, BRT or ferry within 0.5 mi. Weekday and weekend trips 72/30, 132/78, 160/120, 360/216.

OR

PATH 2 · 1–2 PTS

Project-Sponsored Transit Service

Year-round shuttles guaranteed for 3 years. Shuttles = 1, 30 daily trips = 2.

AND/OR

OPT 3 · 1–3 PTS

Walkable Location

Walk Score 60 / 70 / 80, or 4–7 / 8–10 / 11+ uses within 0.5 mi.

AND/OR

OPT 4 · 1–2 PTS · SCHOOLS

Connected Site

Previously developed site. Adjacent, 25% of boundary = 1. Infill, 75% of boundary = 2.

AND/OR

OPT 5 · 1–2 PTS · SCHOOLS

Pedestrian Access

50% / 60% of dwelling units in the attendance boundary within 0.75 mi (grade 8 and below) or 1.5 mi (grade 9 and up). The pathway table lists 1–4.

AND/OR

OPT 6 · 1–2 PTS · DATA CTR, WAREHOUSE

Surrounding Development and Resources

PATH 1 · 1–2 PTS

Development and Adjacency

Former industrial or commercial site = 1. Also bordered by active industrial or commercial parcels = 2.

AND/OR

PATH 2 · 1–2 PTS

Transportation Resources

Two of four = 1, all four = 2. Logistics hub in 10 mi, highway ramp in 1 mi, freight rail access in 1 mi, rail spur.

NUMBERS TO KNOW

0.25 mi	Walk to bus or streetcar stops
0.5 mi	Walk to rail, BRT or ferry
60 / 70 / 80	Walk Score for 1 / 2 / 3 pts
22k / 35k	Surrounding density in sf per acre within 0.25 mi

STANDARDS

Walk Score
Walkability, 60 / 70 / 80

FROM THE PRACTICE TESTS

2 OF 6 QUESTIONS, THE REST ARE IN THE APP

How many uses per type can count, and how many of the five categories must be represented?
Max 2 per type, 3 of 5

What share of the boundary must border previously developed parcels for infill (2 pts) and for adjacent (1 pt)?
75% / 25%

LTC4 · CREDIT

Transportation Demand Management

NC 4 PTS | CS 4 PTS | DECARBONIZATION | QUALITY OF LIFE

Reduce driving and its emissions through parking policy and support for active travel.

REQUIREMENTS

REQ

TDM Assessment

Estimate VMT and baseline versus proposed emissions. Exempt within 0.5 mi of major transit, in transit priority areas or under a mandated TDM program.

AND

OPT 1 · 1–3 PTS

Parking

PATH 1 · 1–3 PTS

Reduce Parking

30% / 60% / 100% below ITE Parking Generation base ratios.

AND/OR

PATH 2 · 2 PTS

Parking Fee

Charge at or above the local market rate.

AND/OR

OPT 2 · 1–3 PTS

Active Travel Facilities

PATH 1 · 1 PTS

Bicycle Network and Storage

Entry or storage within 600 ft of a 3 mi network. Short and long-term storage, at least 4 spaces each.

AND/OR

PATH 2 · 1 PTS

Shower and Changing Facilities

Showers with changing facilities for 1% of regular occupants.

AND/OR

PATH 3 · 1 PTS

Bicycle Maintenance

Permanent, secure repair station with tools and a pump.

NUMBERS TO KNOW

30 / 60 / 100%	Parking reduction for 1 / 2 / 3 pts
1%	Regular occupants served by showers

STANDARDS

ITE Parking Generation Manual, 6th ed.
Base parking ratios

TRANSFERS

→ IPp3
Transportation data is an optional input to the projection.

FROM THE PRACTICE TESTS

9 QUESTIONS

Max distance from an entry or bike storage to the network, and minimum network length?
600 ft / 3 mi

Max walking distance from a functional entry, and minimum spaces per building?
300 ft, at least 4

Which parking is excluded from total provided capacity?
On-street parking in public rights-of-way

What sets the baseline parking ratios?
ITE Parking Generation Manual, 6th edition, or a comparable resource

Which projects are exempt from the Transportation Demand Assessment?
Government-mandated TDM programs, infill affordable housing, and sites in a transit priority area or within 0.5 mi of major transit

What share of regular occupants, and what minimum per dwelling unit?
15%, 1 per 3 units

Where must the repair station be?
In a bike storage area

What is the max posted speed for streets with on-street bike facilities?
25 mph (40 km/h)

What short-term storage does a retail project need?
2 spaces per 5,000 sf, at least 2 per building

LTC5 · CREDIT

Electric Vehicles

NC 2 PTS | CS 2 PTS | DECARBONIZATION

Support electric vehicles to cut tailpipe pollution near communities.

REQUIREMENTS

OPT 1 · 1–2 PTS

Electric Vehicle Supply Equipment

Level 2 or 3, ENERGY STAR connected. Commercial 5% / 10% (min 2 / 4). Residential 10% / 15% (min 5 / 10). One accessible space.

AND/OR

OPT 2 · 1 PTS

Electric Vehicle Readiness

Full 40 A circuits for 10% of commercial spaces (min 10) or 20% residential (min 20).

NUMBERS TO KNOW

5% / 10%	EVSE share of commercial parking for 1 / 2 pts
10% / 20%	EV-ready spaces, commercial / residential

STANDARDS

ENERGY STAR
Efficient appliances and equipment

NFPA 70 (NEC)
Electrical code for Level 2 and 3 chargers

FROM THE PRACTICE TESTS

5 QUESTIONS

A commercial lot has 125 spaces. How many EVSE spaces earn 1 point, and what else is required?
7, one accessible

What infrastructure makes a space EV-ready?
208 to 240 V, 40 A, wired to a junction box

What must every station provide?
Level 2 or 3, 208 to 240 V or greater, ENERGY STAR connected functionality

Can a project with no owned or leased parking pursue the credit?
No NOT IN RS

A commercial project has 80 spaces. How many must be EV-ready?
10



Sustainable Sites

11 points in New Construction · 11 in Core and Shell · the line to The Forest

SSp1	Minimize Site Disturbance	Required
SSc1	Biodiverse Habitat	2 / 2
SSc2	Accessible Outdoor Space	1 / 1
SSc3	Rainwater Management	3 / 3
SSc4	Enhanced Resilient Site Design	2 / 2
SSc5	Heat Island Reduction	2 / 2
SSc6	Light Pollution Reduction	1 / 1

Points shown as NC / CS. A dash means the credit is not offered in that system.

SSP1 · PREREQUISITE

Minimize Site Disturbance

REQUIRED | ECOLOGICAL CONSERVATION AND RESTORATION

Limit construction disturbance and protect native vegetation, healthy soils and habitat.

REQUIREMENTS

REQ

Site Assessment

Conserve 100% of special-status vegetation, fence off healthy plant communities, map and remove invasives.

AND

REQ

Erosion and Sedimentation Control Plan

EPA 2022 CGP regardless of size. Inspect every 7 days, or every 14 days plus within 24 h of a 0.25 in storm.

NUMBERS TO KNOW

7 days	Erosion control inspection interval

STANDARDS

EPA 2022 Construction General Permit
Erosion and sedimentation control

EU Taxonomy DNSH
International alternative to the CGP

TRANSFERS

Landscape line
SSp1 → SSc1 → WEp2 → SSc3 → WEc2 → SSc5

FROM THE PRACTICE TESTS

6 QUESTIONS

What does a zero lot line project with no exterior site disturbance submit?
A narrative NOT IN RS

How often must dewatering be inspected?
Each day dewatering discharges

New plantings must be native or adapted to what?
The EPA Level III ecoregion, or a local equivalent outside the U.S.

Does a site under one acre still follow the 2022 Construction General Permit?
Yes, regardless of size

What can projects outside the U.S. follow instead of the EPA permit?
EU Taxonomy or an equal local standard

How are healthy plant communities protected during construction?
Exclusion zones marked by physical barriers

SSC1 · CREDIT

Biodiverse Habitat

NC 2 PTS | CS 2 PTS | ECOLOGICAL CONSERVATION AND RESTORATION

Conserve natural areas, restore damaged ground and keep birds off the glass.

REQUIREMENTS

OPT 1 · 1–2 PTS

Preserve and Restore Habitat

PATH 1 · 1 PTS

Greenfield Sites

Protect 40% of the greenfield area from all development and construction.

OR

PATH 2 · 1–2 PTS

Previously Disturbed Sites

Restore 20% / 40% of disturbed area (zero lot line 10% / 20%). 10 native species and 110 sf of pollinator habitat.

AND/OR

OPT 2 · 1 PTS

Bird-Friendly Glass

Threat factor 30 maximum from grade to 50 ft, 20 ft above green roofs, and all guardrail glass.

NUMBERS TO KNOW

40%	Greenfield area to preserve
TF 30	Max bird threat factor for glass up to 50 ft

STANDARDS

American Bird Conservancy Threat Factor
Bird collision risk, max 30

EPA Level III ecoregions
Where your 10 native species must come from

7 CFR 657.5, NRCS soil surveys
Prime farmland

TRANSFERS

Landscape line
SSp1 → SSc1 → WEp2 → SSc3 → WEc2 → SSc5

FROM THE PRACTICE TESTS

7 QUESTIONS

How many native or adapted species, and native to what?
10, EPA Level III ecoregion

Minimum total pollinator planting area, and minimum size of each grouping?
110 sf in 10 sf groups

Up to what height from grade, and above a green roof, must glass be assessed?
50 ft from grade, 20 ft above a green roof

What can imported soil not contain?
Prime or important farmland soil, soil from other greenfield sites, or sphagnum peat

Does guardrail glass 60 ft above grade have to comply?
Yes, at any height

What share of the disturbed area must be restored for 1 and 2 points?
20% / 40%

What share must a zero lot line project restore for 1 and 2 points?
10% / 20%

SSC2 · CREDIT

Accessible Outdoor Space

NC 1 PTS | CS 1 PTS | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Provide accessible outdoor space that invites activity and reflects the community it serves.

REQUIREMENTS

REQ
Sufficient Outdoor Space Area
Barrier-free outdoor space of 30% of the site, with 25% of it vegetated.

AND

REQ
Urban Outdoor Space
One of biophilic space, garden, recreation area, or social area seating 5% of occupants.

AND

REQ
Community Outdoor Space
One of daytime public access, two local art pieces, or positive soundscapes near noise sources.

NUMBERS TO KNOW

30% / 25%	Outdoor space share of site, and the vegetated share of that

FROM THE PRACTICE TESTS

2 QUESTIONS

Seating must hold what share of daily average occupants?
5%

What must the outdoor space include?
At least 2 works by local artists

SSC3 · CREDIT

Rainwater Management

NC 3 PTS | CS 3 PTS | ECOLOGICAL CONSERVATION AND RESTORATION

Replicate natural hydrology so runoff does not add to downstream flooding.

REQUIREMENTS

OPT 1 · 1–3 PTS

Percentile of Rainfall Events

Retain the 80th / 85th / 90th percentile event (zero lot line 70 / 75 / 80) with LID and GI. Reuse adds 1 point at the first two tiers.

OR

OPT 2 · 3 PTS

Natural Land Cover Conditions

Retain the runoff increase over natural, predevelopment land cover.

NUMBERS TO KNOW

80 / 85 / 90th	Percentile storm retained for 1 / 2 / 3 pts

TRANSFERS

→ WEc2
Reused rainwater counts as alternative water.

Landscape line
SSp1 → SSc1 → WEp2 → SSc3 → WEc2 → SSc5

FROM THE PRACTICE TESTS

4 QUESTIONS

What must be retained on site?
The increase over natural land cover runoff

How many years of daily rainfall data are needed?
At least 30 years NOT IN RS

Which end uses can the reused rainwater serve?
Irrigation, flushing, makeup, process

Which percentile rainfall events earn 1, 2 and 3 points?
70th / 75th / 80th

SSC4 · CREDIT

Enhanced Resilient Site Design

NC 2 PTS | CS 2 PTS | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Design and maintain the site to withstand the priority hazards found in IPp1.

REQUIREMENTS

REQ · 2 PTS

Two high-priority hazards

Apply the best practices below for at least two of the highest-priority hazards from IPp1.

Drought
Meet WEc2, use native or drought-tolerant planting, water features per SITES C3.4.

AND/OR

Hail
FORTIFIED Commercial hail requirements or local equivalent.

AND/OR

Tsunamis
Designing for Tsunamis siting, TsunamiReady signage and NOAA radios.

AND/OR

Extreme heat
Two of shaded spaces, evaporative cooling, self-shading massing, cooling stations, SR 0.33 paving, open grid.

AND/OR

Hurricanes and high winds
FORTIFIED Commercial Wind or ASCE 7, backup power, storm switch, windbreaks.

AND/OR

Wildfires
NWCG mitigation standards, SITES v2 credit 4.11 and fuel reduction zones.

AND/OR

Flooding
Two strategies per ASCE 24 and FEMA 543. Utilities above DFE, flood-resistant materials, or certification.

AND/OR

Sea level rise
Accommodate projected surge and sea level rise, plus two listed measures.

AND/OR

Winter storms
Two of plow access, snow removal plan, safe heated walks, snow-tolerant planting.

STANDARDS

ASCE 24 and FEMA 543
Flood-resistant design

FORTIFIED Commercial
Hail and hurricane wind

ASCE/SEI 7-10
Wind loads in high-wind zones

SITES (C3.4, v2 credit 4.11)
Water features and wildfire risk

TRANSFERS

← IPp1
The two priority hazards from IPp1 drive SSc4.

Resilience line
IPp1 → SSc4 → EAc6 → EQc4

FROM THE PRACTICE TESTS

7 QUESTIONS

What limit applies to makeup water for a created water feature?
5,000 gal potable or 75% alternative

An emergency cooling station must be within what distance?
0.25 mi

Which references govern?
NWCG Standards and Safer from the Start

Which standard applies?
FORTIFIED Commercial High Wind and Hail Specific Design Requirements

Which wind design standard applies?
ASCE/SEI 7-10

Which references does the credit point to?
NTHMP and TsunamiReady

Which standard applies in hurricane-prone areas?
FORTIFIED Commercial Wind standards

SSC5 · CREDIT

Heat Island Reduction

NC 2 PTS | CS 2 PTS | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Reduce heat islands and their uneven effect on microclimates, habitat and people.

REQUIREMENTS

OPT 1 · 1 PTS

Nonroof and Roof

Nonroof ÷ 0.5 + reflective roof ÷ 0.75 + vegetated roof ÷ 0.75 must cover all paving and roof. Low-slope SRI 82 initial or 64 aged.

AND/OR

OPT 2 · 1 PTS

Parking Under Cover

100% of parking under SRI 32 aged roofing, vegetated roof or energy generation.

AND/OR

OPT 3 · 1 PTS

Tree Equity

Where the Tree Equity Score is high or highest priority, use it to add on-site canopy.

NUMBERS TO KNOW

SRI 82 / 64	Low-slope roof, initial / aged
SR 0.33	Paving reflectance

STANDARDS

ANSI/CRRC S100
Solar reflectance testing

American Forests Tree Equity Score
Canopy equity, U.S. projects only

TRANSFERS

← IPp2
International Tree Equity uses the Human Impact Assessment.

Landscape line
SSp1 → SSc1 → WEp2 → SSc3 → WEc2 → SSc5

FROM THE PRACTICE TESTS

11 QUESTIONS

What share of parking spaces must be under cover?
100%

If one surface meets several heat island strategies, how many times can it count?
Once

What must a covering roof meet if aged SRI data is unavailable?
Initial SRI of at least 39

Do PV canopies over parking count for heat island and renewable energy?
Yes, as shade for SSc5 and as on-site power for EAc4

What must an open-grid pavement system be?
At least 50% unbound

Which U.S. sites qualify, and is keeping the existing canopy enough?
Only Tree Equity Scores of high or highest priority, and canopy must increase

The weighted sum of strategies must be at least what?
Total paving plus total roof area

Which reflectance metric applies to roofs, and which to nonroof hardscape?
SRI roofs, SR hardscape

Which energy generation systems count as shade over paving?
Solar thermal, PV, wind turbines

What must a vegetated roof use?
Native or adapted plant species

Which reflectance values are used?
Aged values, or the initial value where aged data is unavailable

SSC6 · CREDIT

Light Pollution Reduction

NC 1 PTS | CS 1 PTS | ECOLOGICAL CONSERVATION AND RESTORATION

Protect night sky access and limit light spilling onto wildlife and neighbors.

REQUIREMENTS

REQ
Uplight
Uplight rating by MLO zone, U0 in LZ0 and LZ1 up to U4 in LZ4.

AND

REQ
Light Trespass
Backlight and glare ratings per IES TM-15-11 by mounting distance from the lighting boundary.

AND

REQ
Internally Illuminated Exterior Signage
Night luminance cap from 50 cd/m² in LZ0 and LZ1 to 350 cd/m² in LZ4.

STANDARDS

IES TM-15-11 Addendum A, MLO zones
Backlight, uplight and glare (BUG) ratings

FROM THE PRACTICE TESTS

8 QUESTIONS

When is facade and landscape lighting exempt from the uplight and trespass limits?
LZ3 or LZ4, off midnight to 6 a.m.

How far can the lighting boundary move for an abutting street, and for an abutting walkway or plaza?
Street center line / 5 ft NOT IN RS

Can the lighting boundary use the campus property line?
Yes, if contiguous NOT IN RS

What sets a luminaire's maximum backlight rating?
MLO lighting zone, mounting location, and distance from the lighting boundary in mounting heights

In which lighting zones is national flag lighting exempt?
LZ2, LZ3 or LZ4

Does the point require all three of uplight, light trespass and signage limits?
Yes, all three

Name lighting that is exempt from the uplight and trespass limits.
Theatrical, government-mandated roadway, and hospital emergency departments with helipads

Are limits stricter in LZ0 or LZ4?
LZ0, the darkest zone

WE

Water Efficiency

9 points in New Construction · 8 in Core and Shell · the line to The Beach

WEp1	Water Metering and Reporting	Required
WEp2	Minimum Water Efficiency	Required
WEc1	Water Metering and Leak Detection	1 / 1
WEc2	Enhanced Water Efficiency	8 / 7

Points shown as NC / CS. A dash means the credit is not offered in that system.

WEP1 · PREREQUISITE

Water Metering and Reporting

REQUIRED | ECOLOGICAL CONSERVATION AND RESTORATION

Track whole-project water use to manage it and spot further savings.

REQUIREMENTS

REQ

Water metering

Whole-project meters for potable and alternative sources, metered separately. Share data with USGBC yearly for 5 years.

FROM THE PRACTICE TESTS

4 QUESTIONS

A Core and Shell shell has no fixtures or water-using systems in scope. How is it treated?
Exempt from the prerequisite NOT IN RS

Must harvested rainwater be metered separately from municipal water?
Yes

How often, and for how long, must whole-project water data be shared with USGBC?
Annually for 5 years

How can water data be shared with USGBC?
A USGBC data template or an approved third-party source like ENERGY STAR Portfolio Manager NOT IN RS

WEP2 · PREREQUISITE

Minimum Water Efficiency

REQUIRED | DECARBONIZATION | ECOLOGICAL CONSERVATION AND RESTORATION

Set an efficiency floor for fixtures, equipment and irrigation.

REQUIREMENTS

REQ

Minimum Fixture and Fittings Efficiency

OPT 1

Prescriptive Path

Toilet 1.28 gpf, urinal 0.5 gpf, public lav 0.5 gpm, private lav 1.5, kitchen 1.8, shower 2.0.

OR

OPT 2

Performance Path

20% below the fixture and fitting baseline.

AND

REQ

Minimum Equipment Water Efficiency

New appliances meet ENERGY STAR and process tables. No once-through potable cooling.

AND

REQ

Minimum Outdoor Water Efficiency

OPT 1

No Irrigation

No permanent irrigation after a 2-year establishment period.

OR

OPT 2

Efficient Irrigation

Cut the theoretical irrigation requirement 30% through plants and system efficiency.

NUMBERS TO KNOW

1.28 gpf	Toilet
0.5 gpf	Urinal
0.5 gpm	Public lavatory faucet
2.0 gpm	Showerhead
20%	Indoor reduction on the performance path
30%	Irrigation reduction from TIR baseline
2 yr	Max establishment period before no irrigation

STANDARDS

ENERGY STAR
Efficient appliances and equipment

WaterSense
Efficient plumbing fixtures

EPA TIR methodology
Theoretical irrigation requirement baseline

TRANSFERS

→ WEc2
Enhanced water efficiency goes beyond the prerequisite baseline.

Landscape line
SSp1 → SSc1 → WEp2 → SSc3 → WEc2 → SSc5

FROM THE PRACTICE TESTS

5 QUESTIONS

Is once-through cooling with potable water allowed for equipment that rejects heat?
Never

What must an ice machine meet?
ENERGY STAR, air-cooled or closed-loop

Where local supply pressure differs from the LEED baseline, what pressure can flow fixtures be calculated at?
The local standard pressure

What is the default gender mix, and when can it change?
Half male, half female. Changes must hold for the life of the building. NOT IN RS

Which fixtures go in the aggregate calculation?
All new and existing fixtures and fittings in the project boundary

WEC1 · CREDIT

Water Metering and Leak Detection

NC 1 PTS | CS 1 PTS | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Meter subsystems or detect leaks to save water and prevent material damage.

REQUIREMENTS

OPT 1 · 1 PTS

Submeters

Meter 80% of indoor fixtures, irrigation, each makeup system, plus large kitchens and laundries. Hourly data.

CS. Same subsystems plus tenant-ready meters, at least one per floor.

OR

OPT 2 · 1 PTS

Leak Detection Sensors

Sensors on irrigation, 50% of flush fixtures and each makeup system, with alarms and an action plan.

FROM THE PRACTICE TESTS

9 QUESTIONS

Meters must serve at least what share of indoor fixtures and fittings?
80%

What share of flush fixtures need sensors?
50%

Can one sensor on a shared line cover several makeup water systems?
No, one per system

Must residential unit meters be utility owned or utility grade?
No

At how many meals per day must a commercial kitchen be metered?
100

What metering is required for future tenant spaces?
Total potable use, at least 1 per floor

In how many of the listed healthcare subsystems must meters be installed?
Any 5

Does the cooling tower need a submeter under this credit?
No, WEp2 already covers it NOT IN RS

How often must they record, and how is data accessed?
At least hourly, in real time

WEC2 · CREDIT

Enhanced Water Efficiency

NC 8 PTS | CS 7 PTS | DECARBONIZATION | ECOLOGICAL CONSERVATION AND RESTORATION

Cut potable water use further and reward alternative water sources.

> Pursue Option 1 alone, or any mix of Options 2 to 6.

REQUIREMENTS

OPT 1 · 1–8 PTS · CS 1–7

Whole-Project Water Use

30% = 1, then 1 more per 5% to 65% = 8. Alternative water adds a point per tier.

CS. 30% = 1, then 1 more per 5% to 60% = 7.

OR

OPT 2 · 1–3 PTS

Fixture and Fittings

30% / 35% / 40% below the WEp2 calculated baseline.

AND/OR

OPT 3 · 1–2 PTS

Appliance and Process Water

One point per equipment table met. Laundry, kitchen, lab and medical, municipal steam.

AND/OR

OPT 4 · 1–2 PTS · CS 1–3

Outdoor Water Use

PATH 1 · 2 PTS · CS 3

No Irrigation

No permanent irrigation after 2 years of establishment.

OR

PATH 2 · 1–2 PTS · CS 1–3

Efficient Irrigation

TIR cut 50% / 100%.

CS. TIR cut 50% / 75% / 100%.

AND/OR

OPT 5 · 1–2 PTS · CS 1–3

Optimize Process Water Use

PATH 1 · 1–2 PTS · CS 1–3

Limit Cooling Tower Cycles

Max cycles = 1. Plus 25% more cycles or 20% alternative water = 2.

CS. Adds a third tier at 30% more cycles or 30% alternative water.

OR

PATH 2 · 1–2 PTS · CS 1–3

Optimize Water Use for Cooling

25% / 50% less than a water-cooled chiller reference. Baseline system needs a tower.

CS. 25% / 50% / 100% less than the reference.

OR

PATH 3 · 1–2 PTS · CS 1–3

Process Water Use

20% / 30% of process water from alternative sources.

CS. 20% / 30% / 40% from alternative sources. The pathway table lists 1–2.

AND/OR

OPT 6 · 1–2 PTS

Water Reuse

PATH 1 · 1 PTS

Reuse-Ready System

Supply piping and space for future reclaimed or alternative water.

OR

PATH 2 · 2 PTS

Alternative Water Sources

On-site reuse or municipal reclaimed water serving an end use.

Enhanced Water Efficiency CONTINUED

NUMBERS TO KNOW

30 to 65%	Whole-project reduction range for the credit

STANDARDS

ASHRAE 90.1-2019 (addendum cr) or 90.1-2022
Minimum energy efficiency

ENERGY STAR
Efficient appliances and equipment

EPA TIR methodology
Theoretical irrigation requirement baseline

TRANSFERS

← WEp2
Enhanced water efficiency goes beyond the prerequisite baseline.

← SSc3
Reused rainwater counts as alternative water.

Landscape line
SSp1 → SSc1 → WEp2 → SSc3 → WEc2 → SSc5

FROM THE PRACTICE TESTS 9 QUESTIONS

What makes a project eligible?
Appendix G baseline includes a cooling tower

Where does the fixture and fitting baseline come from?
The WEp2 performance path baseline

What earns the second point?
25% more cycles or 20% alternative water

Which equipment categories does it cover?
Washing machines, kitchen equipment, lab and medical equipment, municipal steam

What separates the reuse-ready path from the alternative water path?
Reuse-ready installs the supply system and space for treatment. The other actually uses alternative water.

How much laundry per year must a project process to use the commercial washer table?
120,000 lb per year

What fixture and fitting reduction from the baseline earns the first point?
30%

What TIR reduction is required?
At least 50%

What does alternative water do?
No one-time water analysis needed, and 20% alternative water earns the higher point once max cycles are met

EA

Energy and Atmosphere

33 points in New Construction · **27** in Core and Shell · the line to Power Station

EAp1	Operational Carbon Projection and Decarbonization Plan	Required
EAp2	Minimum Energy Efficiency	Required
EAp3	Fundamental Commissioning	Required
EAp4	Energy Metering and Reporting	Required
EAp5	Fundamental Refrigerant Management	Required
EAc1	Electrification	5 / 4
EAc2	Reduce Peak Thermal Loads	5 / 5
EAc3	Enhanced Energy Efficiency	10 / 7
EAc4	Renewable Energy	5 / 4
EAc5	Enhanced Commissioning	4 / 3
EAc6	Grid Interactive	2 / 2
EAc7	Enhanced Refrigerant Management	2 / 2

Points shown as NC / CS. A dash means the credit is not offered in that system.

EAP1 · PREREQUISITE

Operational Carbon Projection and Decarbonization Plan

REQUIRED | DECARBONIZATION

Show stakeholders the 25-year operational carbon consequences of early design choices.

REQUIREMENTS

REQ
Design Analysis
Test efficiency, peak load and decarbonization measures early with simplified models or comparable data.

AND

REQ
Site Energy Estimate
Annual site energy by fuel type, submitted to USGBC.

AND

REQ
Review Carbon Projection
Owner attests to the 25-year BAU projection, including BPS caps and fines where they apply.

AND

REQ
Decarbonization Plan

PATH 1
Design for Electrification
Earn 4 or 5 points in EAc1.

OR

PATH 2
Plan for Decarbonization
Two-page, 25-year retrofit narrative with timing, cost and readiness measures.

TRANSFERS

→ IPp3
Operational energy data feeds the 25-year carbon projection.

Carbon line
IPp3 → EAp1 → MRp2 → MRc2 → EAc1 → EAc3 → EAc4

FROM THE PRACTICE TESTS

8 QUESTIONS

What does the 25-year BAU projection assume about the grid emissions factor?
Straight-line drop of 95% NOT IN RS

What pattern does the carbon projection reveal?
Grid emissions fall, combustion stays flat NOT IN RS

How many EAc1 points must the project earn?
4 or 5

What does site energy measure?
Energy measured at the building, before generation and delivery losses NOT IN RS

Who must review the carbon projection and decarbonization plan, and what do they do?
Owner or rep attests

Does a smaller, less complex project need an energy model?
No

How can the team estimate annual site energy?
From the EAp2 energy model, or from similar projects or published data

How long can the decarbonization plan narrative be?
No more than 2 pages

EAP2 · PREREQUISITE

Minimum Energy Efficiency

REQUIRED | DECARBONIZATION

Reach a minimum level of energy efficiency under ASHRAE 90.1.

REQUIREMENTS

OPT 1

ASHRAE 90.1-2019

With addendum cr. Only for projects registered before Jan 1, 2028. Appendix G may use future source energy at 2.0.

OR

OPT 2

ASHRAE 90.1-2022

Required from Jan 1, 2028. Appendix G may use future source, site or source energy.

NUMBERS TO KNOW

2019 / 2022	ASHRAE 90.1 editions allowed, with 2022 required from Jan 1, 2028
2.0	Electric site-to-source factor

STANDARDS

ASHRAE 90.1-2019 (addendum cr) or 90.1-2022
Minimum energy efficiency

90.1 Appendix G
Whole-building performance rating method

90.1 Informative Appendix I
Site or source energy as the metric

TRANSFERS

→ EAc3
Option 2 reuses the Appendix G model and the same 90.1 version.

FROM THE PRACTICE TESTS

4 QUESTIONS

Which addendum must be applied with ASHRAE 90.1-2019?
Addendum cr

How does Appendix G differ from the Energy Cost Budget method?
Appendix G uses a fixed baseline NOT IN RS

What limits envelope trade-offs in ECB and Appendix G models?
The building envelope backstop NOT IN RS

What is the proposed design compared to?
A code-minimum version of the same building NOT IN RS

EAP3 · PREREQUISITE

Fundamental Commissioning

REQUIRED | DECARBONIZATION

Verify that systems operate as the owner's project requirements intend.

REQUIREMENTS

REQ

Commissioning requirements

ASHRAE 90.1 Cx with no 4.2.5.2 exceptions. CxP named by end of DD with 2 similar projects. OPR, BOD review, 10% sample, ongoing Cx plan.

NUMBERS TO KNOW

End of DD	CxP named, with 2 similar projects of experience

STANDARDS

ASHRAE 90.1-2019 (addendum cr) or 90.1-2022
Minimum energy efficiency

TRANSFERS

→ EAc5
Enhanced commissioning builds on fundamental commissioning.

← EAc4
Renewable systems are commissioned under EAp3.

FROM THE PRACTICE TESTS

10 QUESTIONS

What is the minimum design meeting attendance for the CxP?
One MEP and one envelope

Which two verification methods qualify?
Air leakage test or air barrier verification NOT IN RS

What is the CxP's role in envelope testing?
Include it in Cx documents and witness a sample NOT IN RS

Can an engineer from the firm that designed the HVAC serve as CxP?
Yes, if not directly involved in designing or installing the systems NOT IN RS

Which 90.1 version governs commissioning?
The same registration-date rule as EAp2

What share of the contractor's completed commissioning documents must the CxP review?
At least 10%

At what construction milestones must Cx meetings happen?
50% and 100% complete

What does a verification and testing provider do?
Carries out functional testing and confirms systems meet their requirements NOT IN RS

Which systems must the OPR include?
HVAC, service water heating, power, lighting, other equipment including on-site renewables, and envelope

What must be developed for the occupancy and operations phase?
An ongoing commissioning plan

EAP4 · PREREQUISITE

Energy Metering and Reporting

REQUIRED | DECARBONIZATION

Track and report energy use and peak demand to find ongoing savings.

REQUIREMENTS

REQ

Energy Monitoring and Recording

Monitoring per ASHRAE 90.1, plus on-site renewable generation.

AND

REQ

Report Energy Data

Monthly energy, generation and peak demand to USGBC every year for 5 years.

STANDARDS

ASHRAE 90.1-2019 (addendum cr) or 90.1-2022
Minimum energy efficiency

TRANSFERS

→ EAc5
Energy metering supports monitoring-based commissioning.

FROM THE PRACTICE TESTS

3 QUESTIONS

Below what share of whole-building electric load can an end use be combined with another?
Less than 10% NOT IN RS

Below what common space area does a residential project count as a small building?
10,000 sf of common space NOT IN RS

How long must the monitoring system store collected data?
36 months NOT IN RS

EAP5 · PREREQUISITE

Fundamental Refrigerant Management

REQUIRED | DECARBONIZATION

Phase out high-GWP refrigerants and reduce leakage.

REQUIREMENTS

OPT 1

No Refrigerants

Use no refrigerants in the project.

OR

OPT 2

Refrigerants

Inventory with GWP, no HCFCs in new equipment, evaluate alternatives above GWP 700, leak check before substantial completion.

NUMBERS TO KNOW

GWP 700	Refrigerant flag for evaluating alternatives, and the HVAC benchmark

TRANSFERS

→ IPp3
Refrigerant data feeds the 25-year carbon projection.

→ EAc7
The refrigerant inventory leads into enhanced management.

FROM THE PRACTICE TESTS

5 QUESTIONS

Which checks are required before charging a system with field-assembled joints?
Leak, vacuum and pressure checks

What basis applies to refrigerant GWP values?
100-year GWP, IPCC Fourth Assessment Report or later

Which refrigerant class is banned in new equipment?
HCFCs, such as R-22 and R-123 NOT IN RS

Does it cap the weighted average GWP of project refrigerants at 700?
No

What happens to refrigerant in removed equipment?
Recover it, then recycle, reclaim or destroy it NOT IN RS

EAC1 · CREDIT

Electrification

NC 5 PTS | CS 4 PTS | PLATINUM GATE | DECARBONIZATION

Design buildings that do not burn fuel on site, for cleaner air and lower carbon as the grid decarbonizes.

Platinum. Platinum requires the full credit, 5 points for NC and 4 for C&S.

Low temperatures means outdoor dry bulb below 20°F. Fossil fuel cells are ineligible.

REQUIREMENTS

OPT 1 · 5 PTS

No On-Site Combustion

No combustion except emergency systems. Heating plus SWH weighted COP of at least 1.8.

OR

OPT 2 · 1–4 PTS

No Combustion Except at Low Temps

PATH 1 · 2 PTS

Space Heating

Combustion only below 20°F. COP 1.8 in climate zones 3 and up.

AND/OR

PATH 2 · 1 PTS

Service Water Heating

COP 1.8 or solar fraction 0.4 when SWH exceeds 34,000 Btu/h.

AND/OR

PATH 3 · 1 PTS

Cooking and Other Process Loads

Cooking, laundry, process and generation run without combustion. Automatic if none exist.

NUMBERS TO KNOW

5 / 4	Points Platinum requires, NC / CS
1.8 COP	Weighted heat pump efficiency
< 20°F	Definition of low temperature

TRANSFERS

Carbon line
IPp3 → EAp1 → MRp2 → MRc2 → EAc1 → EAc3 → EAc4

CORE AND SHELL POINTS

Option 1. Electrification 2–4 · Path 1 3–4 · Path 2 1–3 (Case 1 1–2, Case 2 1) · Path 3 1–2

Option 2. Electrification Readiness 1 · Space, chases, panel capacity and ERV provisions so tenants can install electric heating at COP 1.8.

FROM THE PRACTICE TESTS

4 QUESTIONS

In which climate zones is the 1.8 COP requirement mandatory?
Climate zones 3 and above

What must be provided?
Equipment space, chases, junction boxes, panel capacity, and ERV accommodation for 50% of ventilation air

What weighted COP earns 4 points, and what earns 3?
1.8 / 1.3

How is future tenant heating or hot water equipment counted in the weighted COP?
At COP 1.0, unless code or tenant drawings confirm a weighted 1.8

EAC2 · CREDIT

Reduce Peak Thermal Loads

NC 5 PTS | CS 5 PTS | DECARBONIZATION

Lower peak heating and cooling loads to ease grid demand and make buildings more resilient.

> For every option the envelope must meet ASHRAE 90.1 section 5.5 or 5.6 and cannot be traded off against other systems.

REQUIREMENTS

OPT 1 · 1 PTS

Infiltration and Balanced Ventilation

Supply and exhaust within 10%. Air leakage 0.13 cfm/sf at 50 Pa, 0.20 for major renovation.
cs. Air leakage limit only. Balanced ventilation does not apply to C&S.

AND/OR

OPT 2 · 1 PTS

Ventilation Energy Recovery

70% enthalpy or 75% sensible recovery with bypass. Fan systems under 15% of outdoor air may be excluded.
cs. Fan systems must supply 50% of required ventilation air.

AND/OR

OPT 3 · 1 PTS

Thermal Bridging

ASHRAE 90.1-2022 section 5.5.5(a) prescriptive thermal bridging, with no climate zone 0–3 exception.

AND/OR

OPT 4 · 1–3 PTS

Peak Thermal Load Reductions

Use measured air leakage and balanced ventilation in the load calculation.

PATH 1 · 1–3 PTS

Peak Load Intensity

PHPP or WUFI Passive. Heating plus cooling 16 / 12 / 8 Btu-h/sf, renovation 20 / 15 / 10.

OR

PATH 2 · 1–3 PTS

ASHRAE 90.1 Trade-Off Methods

Appendix C envelope 10% / 20% = 1 / 2. Appendix L ventilation loads 10% = 1.

OR

PATH 3 · 1–3 PTS

Energy Simulation

Appendix G with peak loads as the metric. PI of 0.5 / 0.4 / 0.3.

AND/OR

REQ · 2 PTS · RESIDENTIAL

Option 1 for Residential

Option 1 earns 2 points. Compartmentalize units and test each to 1.5 times the leakage cap.

Reduce Peak Thermal Loads CONTINUED

NUMBERS TO KNOW

0.13 cfm/sf	Air leakage cap at 50 Pa (0.20 for major reno)
10%	Balanced ventilation tolerance
16 / 12 / 8	Peak load in Btu-h/sf for 1 / 2 / 3 pts via PHPP or WUFI

STANDARDS

ASHRAE 62.1-2022
Ventilation and IAQ in commercial spaces

ASTM E779
Whole-building blower door test

ASTM E3158, E1827
Large-building air leakage and orifice blower door

ANSI/RESNET/ICC 380
Dwelling-unit blower door testing

ASHRAE 90.1-2022 Section 5.5.5
Prescriptive thermal bridging

PHPP (PHI) or WUFI Passive (Phius)
Peak heating plus cooling load

ASHRAE 90.1-2019 (addendum cr) or 90.1-2022
Minimum energy efficiency

90.1 Appendix G
Whole-building performance rating method

90.1 Appendix C
Envelope trade-off

90.1 Appendix L
Mechanical system performance (TSPR)

FROM THE PRACTICE TESTS

3 QUESTIONS

What envelope compliance applies to every option?
90.1 Section 5.5 or 5.6

Minimum enthalpy and sensible recovery ratios? Below what share of total outdoor air can fan systems be left out?
70% / 75%, under 15%

Can projects in climate zones 0 to 3 use the 90.1 Section 5.5.5 exceptions?
No

EAC3 · CREDIT

Enhanced Energy Efficiency

NC 10 PTS | CS 7 PTS | PLATINUM GATE | DECARBONIZATION

Minimize energy use beyond the prerequisite to cut extraction, pollution and emissions.

Platinum. Platinum requires 8 points for NC and 5 for C&S.

REQUIREMENTS

OPT 1 · 1–10 PTS

Prescriptive Path

PATH 1 · 1–7 PTS

Regulated Loads

CASE 1 · 1–5 PTS

ASHRAE 90.1-2019

Section 11 credits. 25 / 50 / 75 / 100 / 125 = 1 to 5. Before Jan 1, 2028 only.

OR

CASE 2 · 4–7 PTS

ASHRAE 90.1-2022

The 90.1-2022 minimum = 4, then +25 / +50 / +75 credits = 5 / 6 / 7.

AND/OR

PATH 2 · 1–4 PTS

Plug and Process Loads

CASE 1 · 1 PTS

Plug Load Management

Occupant plug load dashboard and IT shutoff policies.

AND/OR

CASE 2 · 1–4 PTS

Efficient PPL Equipment

90% of equipment meets criteria. 1 / 2 / 3+ categories = 1 / 2 / 3. Process-intensive 0.5 / 1.0 W/sf = 3 / 4.

OR

CASE 3 · 1–4 PTS

PPL Exceptional Calculation

G2.5 method. 10% / 20% / 30% / 40% improvement.

OR

OPT 2 · 1–10 PTS

Energy Simulation

Appendix G with future source energy and LEED BPFs.

PATH 1 · 1–10 PTS

Excluding On-Site Renewables

3% below PIt = 1, then 1 more per 3% to 30% = 10.

OR

PATH 2 · 1–10 PTS

Including On-Site Renewables

10% below PIt = 1, then 1 more per 10% to 100% = 10.

CORE AND SHELL POINTS

Option 1. Prescriptive Path 1–7 · Path 1 1–6 (Case 1 1–5, Case 2 2–6) · Path 2 1–3 (Case 1 1, Case 2 1–3)

Option 2. Energy Simulation 1–7 · Path 1 1–7 · Path 2 1–7

Enhanced Energy Efficiency CONTINUED

NUMBERS TO KNOW

8 / 5	Points Platinum requires, NC / CS

STANDARDS

ASHRAE 90.1-2019 (addendum cr) or 90.1-2022
Minimum energy efficiency

90.1 Appendix G
Whole-building performance rating method

ASHRAE 90.4-2022
Data center electrical efficiency

ISO 25745
Energy class of elevators, escalators and walkways

ENERGY STAR
Efficient appliances and equipment

California Title 24-2022, Section 120.6
Refrigerated warehouses

TRANSFERS

← EAp2
Option 2 reuses the Appendix G model and the same 90.1 version.

Carbon line
IPp3 → EAp1 → MRp2 → MRc2 → EAc1 → EAc3 → EAc4

FROM THE PRACTICE TESTS

11 QUESTIONS

What reduction below the PI target earns 10 points, excluding and including on-site renewables?
30% / 100%

What two things does the point require?
Dashboard and IT shutoff policy

What share of each claimed category must comply?
90% by quantity or rated load

What must elevators and escalators meet?
ISO 25745, at least Class A

What is Case 3, and can it combine with Cases 1 and 2?
Exceptional calculation for plug and process loads, pursued on its own

Can the regulated loads and plug and process loads paths be combined?
Yes, up to the option maximum

Give examples of process, or unregulated, energy.
Computers, kitchen, laundry, elevators NOT IN RS

Which version of ASHRAE 90.1 does it use?
The version used for EAp2

What do Path 1 and Path 2 cover?
Regulated loads / plug and process loads

What is nonprocess, or regulated, energy?
Systems that condition spaces and provide comfort, like lighting, HVAC, hot water, toilet exhaust and garage ventilation NOT IN RS

Which 90.1 Section 11.5.2 measure can count toward points, night flush or on-site renewables?
Building mass with night flush (G07)

EAC4 · CREDIT

Renewable Energy

NC 5 PTS | CS 4 PTS | PLATINUM GATE | DECARBONIZATION

Supply or procure renewable energy, with new and on-site generation weighted most.

Platinum. Platinum requires 100% of site energy (base building energy for C&S) from any mix of tiers.

Contracts count at 10 years, prorated if shorter. EACs must be retired for the project and generated no earlier than 18 months before submission.

REQUIREMENTS

OPT 1 · 1–5 PTS · CS 1–4

Renewable Energy Supply or Procurement

Add tiers together up to the maximum.

TIER 1 · 1–5 PTS · CS 1–4

On-Site or Social Impact

1 W/sf or 5% = 1, 2 W/sf or 10% = 2, 20% = 3, 35% = 4, 100% Tier 1 or 2 = 5.

CS. Of base building energy. 1 W/sf or 15% = 1, 2 W/sf or 30% = 2, 65% = 3, 100% = 4.

AND/OR

TIER 2 · 1–5 PTS · CS 1–4

New Off-Site Electricity

New assets. 20% / 40% / 60% / 80% = 1 to 4, 100% Tier 1 or 2 = 5.

CS. 35% / 70% / 100% / 200% = 1 to 4.

AND/OR

TIER 3 · 1–2 PTS

Off-Site Renewable Energy

Green-e electricity or fuels. 50% = 1, 100% = 2.

CS. 100% = 1, 200% = 2.

AND/OR

OPT 2 · CS 1 · CS ONLY

Renewable Energy Readiness

Solar zone of 40% of roof, mounting plan, conduit pathway, 200 A busbar with a reserved breaker.

NUMBERS TO KNOW

100%	Renewable share of energy use Platinum requires
10 yr	Renewable energy contract length

STANDARDS

Green-e Energy, Green-e Renewable Fuels
Certification of off-site renewables

TRANSFERS

→ EAp3
Renewable systems are commissioned under EAp3.

Carbon line
IPp3 → EAp1 → MRp2 → MRc2 → EAc1 → EAc3 → EAc4

FROM THE PRACTICE TESTS

4 OF 12 QUESTIONS, THE REST ARE IN THE APP

Where can Tier 1 renewable equipment be located?
Site, campus or social impact project

Does a vapor-compression geothermal heat pump count as renewable energy?
No NOT IN RS

The solar zone must equal what share of the gross roof area?
40%

What does the main electrical panel need?
200 A busbar, reserved breaker space

EAC5 · CREDIT

Enhanced Commissioning

NC 4 PTS | CS 3 PTS | DECARBONIZATION

Make sure systems and enclosure work as designed and keep performing over time.

REQUIREMENTS

OPT 1 · 1–3 PTS

Enhanced Commissioning

Independent CxP named in predesign or very early design.

PATH 1 · 2 PTS

MEP Systems

ASHRAE 202-2024 for MEP, controls, process and renewables. Two design meetings, training verified.

AND/OR

PATH 2 · 1 PTS

Building Enclosure

ASTM E2947-21a BECx with air leakage, water penetration and infrared testing.

AND/OR

OPT 2 · 1–2 PTS

Monitoring-Based Commissioning

PATH 1 · 1 PTS

Basic Software

3-year MBCx contract and an energy information system with hourly data.

OR

PATH 2 · 2 PTS

Enhanced Software

Path 1 plus FDD above 600 tons, anomaly alerts, normalization and GHG reporting.

CORE AND SHELL POINTS

Option 1. Enhanced Commissioning 1–2 · Path 1 1 · Path 2 2

Option 2. Monitoring-Based Commissioning 1 · 3-year MBCx and an EIS expandable to tenants, with a tenant portal for spaces over 10,000 sf.

NUMBERS TO KNOW

3 yr	Monitoring-based commissioning commitment

STANDARDS

ASTM E2947-21a
Building enclosure commissioning

ASTM E779
Whole-building blower door test

ASTM E3158, E1827
Large-building air leakage and orifice blower door

ASTM E783, E1186
Air leakage at installed windows and leak location

ASTM E1105, AAMA 501.2
Field water penetration testing

ASTM C1060, C1153
Infrared imaging of insulation and wet roof insulation

ANSI/ASHRAE/IES 202-2024
The commissioning process

TRANSFERS

← EAp3
Enhanced commissioning builds on fundamental commissioning.

← EAp4
Energy metering supports monitoring-based commissioning.

EAC6 · CREDIT

Grid Interactive

NC 2 PTS | CS 2 PTS | DECARBONIZATION

Make the building an active, flexible partner in grid reliability and decarbonization.

Every project first evaluates grid-interactive measures and provides interval meters and equipment that accepts an external signal.

REQUIREMENTS

OPT 1 · 1–2 PTS

Energy Storage

Electric 0.2 / 0.4 kWh per kW of peak, or thermal 1.0 / 2.0 per unit of peak thermal demand, with automatic shifting.

AND/OR

OPT 2 · 1 PTS

Demand Response Program

At least a 1-year DR contract with intent to renew. On-site generation and combustion do not count.

AND/OR

OPT 3 · 1 PTS

Automated Demand-Side Management

PATH 1 · 1 PTS

System-Level Controls

Automated DR on two systems. HVAC 50%, lighting 50%, receptacles, SWH 90% or EVSE.

OR

PATH 2 · 1 PTS

Building Automation System

Automatically shed 10% of peak demand for 1 hour on grid signals, winter and summer.

AND/OR

OPT 4 · 1 PTS

Power Resilience

Island critical loads on on-site renewables and storage for 3 days.

NUMBERS TO KNOW

0.2 / 0.4	Electric storage in kWh per kW of peak demand
3 days	Islanding of critical loads

TRANSFERS

Resilience line
IPp1 → SSc4 → EAc6 → EQc4

FROM THE PRACTICE TESTS

7 QUESTIONS

How must the energy storage operate?
Charge off-peak, discharge on-peak

Automated demand response must cover how many eligible systems?
At least 2

How much peak demand must the plan shed, and for how long?
At least 10% for at least 1 hour

What coverage applies to HVAC, lighting and service water heating?
HVAC 50% of capacity, lighting 50% of power, SWH 90% of capacity

Can on-site generation or fuel combustion, renewables included, meet the demand-side management criteria?
No

What demand response contract is required?
At least 1 year, intending renewal

What applies under every option?
Evaluate against current and forecast grid context, plus interval meters and equipment that accepts an external signal

EAC7 · CREDIT

Enhanced Refrigerant Management

NC 2 PTS | CS 2 PTS | DECARBONIZATION

Speed the shift to low-GWP refrigerants and manage leakage better.

REQUIREMENTS

OPT 1 · 1–2 PTS

No Refrigerants or Low GWP

PATH 1 · 1 PTS

No Refrigerants

No refrigerant-containing equipment in the project.

cs. Not available if future equipment will serve heating or cooling.

OR

PATH 2 · 1–2 PTS

Low GWP Refrigerants

Weighted GWP at 80% / 50% of benchmarks. HVAC 700, heat pump water heaters 1,400, other process 300.

AND/OR

OPT 2 · 1 PTS

Limit Refrigerant Leakage

Self-contained equipment above GWP 700, leak detection above 100 tCO2e, brazed fittings and a maintenance plan.

cs. Needs 20% of heating and cooling capacity in scope. Same design, install and operation rules.

AND/OR

OPT 3 · 1–2 PTS

GreenChill Certification

Food retail over 20% of area. Silver = 1, Gold or Platinum = 2.

STANDARDS

EPA GreenChill
Supermarket refrigerant management

TRANSFERS

← EAp5
The refrigerant inventory leads into enhanced management.

FROM THE PRACTICE TESTS

9 QUESTIONS

When must equipment be self-contained with no field-installed piping?
Above 700 GWP, and 80% of total GWP

When is an automatic leak detection system required?
Enclosed spaces over 100 tCO2e charge

What operations document is required?
A refrigerant maintenance plan

Which projects are eligible?
Projects where food retailing is more than 20% of gross area

What does it require?
No refrigerant-containing equipment

On what basis is refrigerant GWP compared to the benchmarks?
Weighted average of all new equipment

How must weighted average GWP compare to the benchmark for 1 and 2 points?
80% / 50% or less

Weighted average GWP must stay at or under what share of the benchmark for 1 and 2 points?
80% / 50%

What share of heating and cooling capacity must be in scope?
At least 20%

MR

Materials and Resources

18 points in New Construction · 21 in Core and Shell · the line to The Market

MRp1	Planning for Zero Waste Operations	Required
MRp2	Quantify and Assess Embodied Carbon	Required
MRc1	Building and Materials Reuse	3 / 5
MRc2	Reduce Embodied Carbon	6 / 8
MRc3	Low-Emitting Materials	2 / 1
MRc4	Building Product Selection and Procurement	5 / 5
MRc5	Construction and Demolition Waste Diversion	2 / 2

Points shown as NC / CS. A dash means the credit is not offered in that system.

MRP1 · PREREQUISITE

Planning for Zero Waste Operations

REQUIRED | DECARBONIZATION | ECOLOGICAL CONSERVATION AND RESTORATION

Set the building up for zero waste operations with space for recycling and organics.

REQUIREMENTS

REQ

Storage and Collection of Recyclables

Space for organics, paper, cardboard, glass, plastics and metals, plus batteries, lamps and e-waste.

AND

REQ

Zero Waste Operations Planning

Design details and manuals that help operators reach high waste prevention goals.

FROM THE PRACTICE TESTS

6 QUESTIONS

What must every project provide?
Dedicated recycling collection and storage areas

List the true waste hierarchy from most to least preferred.
Reduce, reuse, recycle, compost, digest NOT IN RS

Which phase does MRp1 address, and which credit covers construction waste?
Operations. MRc5 covers construction waste.

If no organics collection exists at occupancy, is organics storage space still required?
Yes

With commingled recycling allowed and no organics service, which stream still needs dedicated storage?
Organics and food waste

What extra step is required?
Share the waste infrastructure and service options in the tenant guidelines

MRP2 · PREREQUISITE

Quantify and Assess Embodied Carbon

REQUIRED | DECARBONIZATION

Quantify cradle-to-gate embodied carbon and find its largest sources.

REQUIREMENTS

REQ

Embodied Carbon

A1 to A3 GWP for structure, enclosure and hardscape, ancillary structures included.

AND

REQ

High-Priority Embodied Carbon Sources

Name the top three hot spots and the strategies considered for each.

NUMBERS TO KNOW

A1 to A3	Embodied carbon scope for the prerequisite
Top 3	Embodied carbon hot spots to identify

TRANSFERS

→ IPp3
Embodied carbon data feeds the 25-year carbon projection.

→ MRc2
The embodied carbon inventory grows into reduction.

Carbon line
IPp3 → EAp1 → MRp2 → MRc2 → EAc1 → EAc3 → EAc4

FROM THE PRACTICE TESTS

4 QUESTIONS

Must a detached garage and outbuilding inside the project boundary be included?
Yes, include them

What A1 to A3 GWP do salvaged steel beams get?
Zero NOT IN RS

Which materials must the embodied carbon calculation include at a minimum?
Asphalt, concrete, masonry, structural steel, insulation, aluminum extrusions, structural wood and composites, cladding, glass

What extra step is required?
Share embodied carbon results and material suppliers in the tenant guidelines

MRC1 · CREDIT

Building and Materials Reuse

NC 3 PTS | CS 5 PTS | DECARBONIZATION | ECOLOGICAL CONSERVATION AND RESTORATION

Reuse buildings and materials to keep carbon, resources and history in place.

REQUIREMENTS

OPT 1 · 1–3 PTS · CS 1–5

Building Reuse

Existing structure and enclosure by project area. 20% / 35% / 50%.
cs. 10% / 20% / 30% / 40% / 50% = 1 to 5.

AND/OR

OPT 2 · 1–2 PTS

Materials Reuse

15% of 1 targeted or 2 other types = 1. 30% of 1 targeted, 15% of 2 targeted or 15% of 4 other = 2.

NUMBERS TO KNOW

20 / 35 / 50%	Building reuse for 1 / 2 / 3 pts, NC

TRANSFERS

→ MRc5
Salvage sent off site for reuse counts toward C&D diversion.

FROM THE PRACTICE TESTS

11 QUESTIONS

Which four materials are targeted?
Carpet, ceilings, furniture, interior walls

Which elements count as maintained structure?
Floor decking, roof decking, enclosure

Do recycled content, surplus or overstock count as reused material?
No NOT IN RS

Who should be involved?
Salvage experts such as deconstruction contractors NOT IN RS

With an addition, what floor area is the reuse calculation based on?
The existing floor area only

Does salvaged brick bought from a reuse market count?
Yes, kept-in-place and purchased salvage both count

Does salvaged structural steel bought from another site count, and under which option?
Yes, under Option 2 Materials Reuse

How can one targeted material type earn 2 points?
Reuse at least 30% of it

What units measure reused doors, carpet and dimensional lumber?
Count / surface area / board foot

Where do salvaged materials count, by destination?
Kept on site counts for MRc1. Sent off site for reuse counts for MRc5.

How are structurally unsound or hazardous portions treated?
Excluded from the calculations entirely

MRC2 · CREDIT

Reduce Embodied Carbon

NC 6 PTS | CS 8 PTS | PLATINUM GATE | DECARBONIZATION

Track and reduce embodied carbon in structure, enclosure and hardscape materials.

Platinum. Platinum requires a 20% embodied carbon reduction.

REQUIREMENTS

OPT 1 · 1–6 PTS · CS 1–7

Whole-Building LCA

Modules A to C. Meet baseline = 2, then 10% / 20% / 30% / 40% = 3 to 6.
CS. Adds 50% = 7.

AND/OR

OPT 2 · 1–3 PTS · CS 1–4

EPD Analysis

PATH 1 · 1–3 PTS · CS 1–4

Project-Average Approach

Industry average = 1, 20% = 2, 40% = 3.
CS. Adds 50% = 4.

OR

PATH 2 · 1–2 PTS · CS 1–3

Materials-Type Approach

Beat benchmarks in 3 categories = 1, 5 or more = 2.
CS. 2 / 4 / 6 categories = 1 / 2 / 3.

AND/OR

OPT 3 · 1–2 PTS

Track Construction Emissions

PATH 1 · 1 PTS

Contractor

Track A5 fuel and utility use for contractor jobsite operations.

OR

PATH 2 · 2 PTS

Contractor and Subcontractors

Track A5 for contractor and subcontractor operations.

NUMBERS TO KNOW

20%	Embodied carbon reduction Platinum requires
A to C	Whole-building LCA scope
10/20/30/40%	WBLCA reductions for 3 to 6 pts

STANDARDS

Type III EPDs, EPA or CLF Material Baselines
Product carbon data and industry averages

TRANSFERS

← MRp2
The embodied carbon inventory grows into reduction.

Carbon line
IPp3 → EAp1 → MRp2 → MRc2 → EAc1 → EAc3 → EAc4

FROM THE PRACTICE TESTS

6 OF 10 QUESTIONS, THE REST ARE IN THE APP

What service life must the baseline and proposed share, and how much can quantities change?
60 years / 10%

What counts as one qualifying category?
Each material type beating its benchmark

For salvaged materials, which stages are zeroed and which stay in?
Zero A1 to A3, keep A4 and B to C NOT IN RS

What kind of EPDs do both paths require?
Type III, product-specific

What does it reward?
Tracking construction-phase emissions

Baseline and proposed match in size, orientation and energy performance. What can differ?
The contractor NOT IN RS

MRC3 · CREDIT

Low-Emitting Materials

NC 2 PTS | CS 1 PTS | QUALITY OF LIFE

Specify products that keep VOCs and formaldehyde out of indoor air.

REQUIREMENTS

PATH 1 · 1 PTS

Core Three Categories

Paints and coatings, flooring and ceilings, each over 90%.

OR

PATH 2 · 2 PTS

Core Three plus Two

Path 1 plus two of adhesives, walls, insulation and composite wood, each over 80%.

OR

PATH 3 · 2 PTS

Core Three plus Furniture

Path 1 plus furniture over 80%.

CORE AND SHELL POINTS

Any Three Categories 1 · Any three of the seven categories, each over 90%. Tenant-installed products excluded.

NUMBERS TO KNOW

90% / 80%	Low-emitting thresholds, core three categories / additional

STANDARDS

CDPH Standard Method v1.2-2017
VOC emissions testing of products

ANSI/BIFMA M7.1-2011 (R2021), e3-2024
Furniture emissions and sustainability

EPA TSCA Title VI, CARB ATCM
Formaldehyde in composite wood (ULEF or NAF)

FROM THE PRACTICE TESTS

13 QUESTIONS

Which categories do interior doors, countertops and spray foam insulation fall under?
Walls / furniture / insulation

A lab report must be dated no more than how long before purchase?
3 years

How does furniture earn the second point?
Path 3, furniture over 80%

How are tenant-installed products handled?
Excluded, base list in tenant guidelines

How are salvaged or reused materials treated for VOC emissions?
They comply automatically

A separately bought finish is applied off-site to stone. How is it documented?
As a separate product under the low-emitting criteria NOT IN RS

Which products are inherently non-emitting?
Stone, ceramic, powder-coated or plated metal, untreated solid wood NOT IN RS

Name products that never count in any low-emitting category.
Poured concrete, structure, MEP

What does it include, and what falls elsewhere or is excluded?
Walk-off, rugs, raised floors, wood subfloor

Which three categories must exceed 90%?
Paints and coatings, flooring, ceilings

What formaldehyde standard does LEED apply?
ULEF or NAF resins

Which test method and scenario apply to a qualifying lab report?
CDPH Standard Method v1.2-2017, private office scenario

Which insulation is excluded?
Insulation installed outside an air barrier

MRC4 · CREDIT

Building Product Selection and Procurement

NC 5 PTS | CS 5 PTS | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Choose products with disclosed, multi-attribute sustainability performance.

Criteria areas are climate health, human health, ecosystem health, social health and equity, and circular economy. Levels score 1x, 2x and 3x, capped at 5 per product.

REQUIREMENTS

REQ · 1–5 PTS

Product categories

Each category whose MAS-adjusted value exceeds 100% earns 1 point.

1 PTS
Paints and coatings

AND/OR

1 PTS
Walls

AND/OR

1 PTS
Furniture

AND/OR

1 PTS
Adhesives and sealants

AND/OR

1 PTS
Ceilings

AND/OR

1 PTS
Composite wood

AND/OR

1 PTS
Flooring

AND/OR

1 PTS
Insulation

AND/OR

1 PTS
Plumbing fixtures

STANDARDS

Mindful Materials framework, AIA Materials Pledge
The five health and impact criteria

FROM THE PRACTICE TESTS

10 QUESTIONS

What earns Level 1?
Quantified, disclosed embodied carbon NOT IN RS

Which industry frameworks do the product criteria align with?
Mindful Materials and AIA Pledge NOT IN RS

What earns Level 1?
Recycled or bio-based content NOT IN RS

Which product categories count, and are plumbing fixtures one?
9 categories, plumbing included

What is a product called that achieves in two or more criteria areas?
A multi-attribute product

Name the five criteria areas.
Climate, human, ecosystem, social, circular

Describe Levels 1, 2 and 3 and give their multipliers.
1× / 2× / 3×

A product earns Level 3 in one area and Level 2 in another. What is its multi-attribute score?
5

What adjusted value must a product category exceed to earn a point?
100%

A product has two documents in the same criteria area. How do they score?
Only the highest score in that area counts NOT IN RS

MRC5 · CREDIT

Construction and Demolition Waste Diversion

NC 2 PTS | CS 2 PTS | DECARBONIZATION | ECOLOGICAL CONSERVATION AND RESTORATION

Keep construction and demolition waste out of landfills and incinerators.

REQUIREMENTS

REQ

C&D Materials Management Plan

Develop and follow the plan, then report all waste, disposal and diversion.

AND

REQ · 1–2 PTS

Diversion

50% with 10% salvaged or source-separated = 1. 75% with 25% = 2. Salvage counts 200%, unverified mixed caps at 35%.

NUMBERS TO KNOW

50% / 75%	C&D diversion, with 10% / 25% salvaged or source-separated
2x	Diversion credit for salvaged materials
35%	Cap on unverified mixed recycling

TRANSFERS

← MRc1
Salvage sent off site for reuse counts toward C&D diversion.

FROM THE PRACTICE TESTS

7 QUESTIONS

Does burning recovered wood from recycling (biomass) count as diversion?
No, it counts as 0% diverted

What must flow to future tenants?
The C&D waste management plan

What earns the first point?
50% diverted, with 10% of it salvaged or source-separated

Metal separated on site goes to a mixed-material facility. How does it count?
As commingled, at the facility average rate NOT IN RS

How are materials salvaged and reused on site handled?
Excluded, they count under MRc1

Which materials are excluded from the diversion calculation?
Excavated soil and land-clearing debris

How are alternative daily cover and incineration counted?
Both are waste, 0% diverted

EQ

Indoor Environmental Quality

13 points in New Construction · 11 in Core and Shell · the line to Home

EQp1	Construction Management	Required
EQp2	Fundamental Air Quality	Required
EQp3	No Smoking or Vehicle Idling	Required
EQc1	Enhanced Air Quality	1 / 1
EQc2	Occupant Experience	7 / 7
EQc3	Accessibility and Inclusion	1 / 1
EQc4	Resilient Spaces	2 / 2
EQc5	Air Quality Testing and Monitoring	2 / –

Points shown as NC / CS. A dash means the credit is not offered in that system.

EQP1 · PREREQUISITE

Construction Management

REQUIRED | QUALITY OF LIFE

Protect workers and future occupants from construction air quality problems.

REQUIREMENTS

REQ

Construction management practices

No smoking within 25 ft, extreme heat protection, HVAC protection, source control, pathway interruption, housekeeping, scheduling.

NUMBERS TO KNOW

25 ft	Smoking setback from entries, intakes and windows

FROM THE PRACTICE TESTS

9 QUESTIONS

If permanent HVAC must run during construction, what is required?
Maintain proper filtration

Which guideline are the construction management practices adapted from?
SMACNA IAQ Guidelines, Ch. 3 NOT IN RS

Which stored materials must be protected from moisture?
Ceiling panels, carpet, fabric, insulation

Whose guidance is it adapted from?
OSHA heat-prevention guidance NOT IN RS

What does the plan require?
Vacuums with high-efficiency particulate filters, and sweeping compounds or wetting agents

In what order do filter replacement and testing and balancing happen?
New filters first, then TAB

How long must the no-smoking signage stay up, and what does the ban cover?
Until construction is complete

Is a preoccupancy flush-out required?
No

When do carpet and ceiling tile go in relative to paint?
After wet-applied materials are fully cured and dry NOT IN RS

EQP2 · PREREQUISITE

Fundamental Air Quality

REQUIRED | QUALITY OF LIFE

Design ventilation and filtration for above-average indoor air quality.

REQUIREMENTS

REQ

Investigate Regional and Local Air Quality

ASHRAE 62.1-2022 sections 4.1 to 4.3.

AND

REQ

Ventilation and Filtration Design

62.1-2022 sections 5 and 6. MERV 13, ePM1 50% or tested in-room air cleaning. Sealed or power-vented combustion.

AND

REQ

Entryway System Design

Permanent entryway systems at primary exterior entrances. No length requirement.

STANDARDS

ASHRAE 62.1-2022
Ventilation and IAQ in commercial spaces

ASHRAE 62.1-2022 Addendum N
Testing in-room air cleaners for particles and gases

ASHRAE 62.2-2022
Ventilation and local exhaust in dwelling units

ASHRAE 52.2-2017
Lab test that assigns MERV ratings to filters

ISO 16890-2016
International filter classes

ASHRAE 241-2023
Control of infectious aerosols

ASHRAE 170-2021
Ventilation of healthcare facilities

ENERGY STAR H/ERV spec v2.3, HVI certified
Residential HRVs, ERVs and bath fans

TRANSFERS

→ EQc1
Enhanced air quality exceeds the EQp2 design.

FROM THE PRACTICE TESTS

9 QUESTIONS

Which standards define acceptable indoor air quality?
ASHRAE 62.1-2022 and 62.2-2022

Above what outdoor air intake must a system have airflow measurement?
1,000 cfm

Which ASHRAE 62.1 procedures can be used?
Ventilation rate, IAQ, natural ventilation

Where must CO monitors go?
On each floor of every dwelling unit, hard-wired with battery backup

Does a recirculating range hood meet the kitchen exhaust requirement?
No, exhaust must be ducted outdoors

What minimum filtration does a central HVAC system need?
MERV 13

Which standard rates filtration (MERV)?
ASHRAE 52.2-2017

What must an indoor fireplace or woodstove have?
Solid glass enclosures or doors that seal when closed

Where are entryway systems required, and how long must they be?
Primary exterior entrances, with no length requirement



EQP3 · PREREQUISITE

No Smoking or Vehicle Idling

REQUIRED | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Keep tobacco smoke, cannabis smoke and vehicle exhaust away from people.

REQUIREMENTS

REQ

Prohibit Smoking

Indoors, and outdoors except 25 ft from entries, intakes and operable windows. Schools ban it on site.

AND

REQ

Prohibit Vehicle Idling

No idling on site, with signage or enforcement.

STANDARDS

ANSI/RESNET/ICC 380
Dwelling-unit blower door testing

FROM THE PRACTICE TESTS

4 QUESTIONS

What is the smoking rule, and where does the signage go?
All smoking banned, signs on property line

Does the smoking ban cover e-cigarettes and cannabis?
Yes

What is the vehicle rule, and how is it communicated?
No idling on site, with signage or enforcement

When can smoking be allowed inside dwelling units?
When each smoking unit is compartmentalized, tested to 1.5 times the table leakage limits

EQC1 · CREDIT

Enhanced Air Quality

NC 1 PTS | CS 1 PTS | QUALITY OF LIFE

Exceed ASHRAE 62.1 to better protect occupant health.

REQUIREMENTS

OPT 1 · 1 PTS

Increased Ventilation

15% above minimum outdoor air in 95% of occupied spaces. 30% is exemplary.

OR

OPT 2 · 1 PTS

Enhanced IAQ Design

PM2.5 10 µg/m³, formaldehyde 20 µg/m³, ozone 10 ppb. The required route for IAQP projects.

NUMBERS TO KNOW

+15%	Ventilation boost to 95% of occupied spaces (30% exemplary)
10 / 20 / 10	PM2.5 µg/m³ / formaldehyde µg/m³ / ozone ppb

STANDARDS

ASHRAE 62.1-2022
Ventilation and IAQ in commercial spaces

TRANSFERS

← EQp2
Enhanced air quality exceeds the EQp2 design.

FROM THE PRACTICE TESTS

5 QUESTIONS

A project used the IAQP for EQp2. Which option can it use?
Option 2 only

Which standard do both options build on?
ASHRAE 62.1-2022, exceeding Section 6

Which reference sets the design compounds and limits?
ASHRAE 62.1-2022, Tables 6-5 and 6-6, with lower enhanced limits

Match the ASHRAE numbers. Which covers ventilation, filters, respiratory disease air cleaning, and the smoke-ready mode?
62.1 ventilation, 52.2 filters (MERV), 241 air cleaning, Guideline 44 smoke mode

How much more outdoor air, and delivered to how much of the building?
15% more, to 95% of regularly occupied spaces

EQC2 · CREDIT

Occupant Experience

NC 7 PTS | CS 7 PTS | QUALITY OF LIFE

Move past neutral spaces toward ones people enjoy, adjust and feel connected to.

REQUIREMENTS

OPT 1 · 1–4 PTS

Biophilic Environment

PATH 1 · 1 PTS

Integrated Biophilic Design

Demonstrate all five Kellert and Calabrese principles.

AND/OR

PATH 2 · 2–3 PTS

Quality Views

Views for 75% / 90% of occupied area through glass with VLT over 40%. Atria count for up to 30%.

AND/OR

OPT 2 · 1 PTS

Adaptable Environment

Give occupants choice across thermal, sound and lighting zones.

AND/OR

OPT 3 · 1 PTS

Thermal Environment

AND/OR

OPT 4 · 1–2 PTS

Sound Environment

PATH 1 · 1 PTS

Mapping Acoustic Expectations

Map acoustic expectations for indoor and outdoor spaces.

OR

PATH 2 · 2 PTS

Acoustic Criteria

Meet acoustic criteria for indoor and outdoor spaces.

AND/OR

OPT 5 · 1–6 PTS

Lighting Environment

PATH 1 · 1 PTS

Solar Glare

AND/OR

PATH 2 · 1 PTS

Quality Electric Lighting

UGR 19 or lower, and CRI 90 or TM-30 Rf 78 and Rg 95.

AND/OR

PATH 3 · 1 PTS

Proximity to Windows

30% of occupied area within 20 ft of glazing with VLT over 40%.

OR

PATH 4 · 1–4 PTS

Daylight Simulation

Average sDA 300/50% of 40% / 55% / 65% / 75%. Address spaces with ASE over 20%.

CORE AND SHELL POINTS

Option 1. Biophilic Environment	1–4 · Path 1 1 · Path 2 2–3 · Path 3 1
Option 2. Lighting Environment	1–6 · Path 1 1 · Path 2 1 · Path 3 1 · Path 4 1–4

Occupant Experience CONTINUED

NUMBERS TO KNOW

40/55/65/75%	Daylight sDA for 1 to 4 pts
30% in 20 ft	Occupied area near glazing with VLT over 40%

STANDARDS

ASHRAE 55-2023
Thermal comfort

IES LM-83-23
Daylight metrics sDA 300/50% and ASE 1000,250

IES TM-30-20
Color rendering, Rf 78 and Rg 95

NEMA UGR white paper
Luminaire glare, UGR 19 or lower

Kellert and Calabrese
Five biophilic design principles

FROM THE PRACTICE TESTS

19 QUESTIONS

What does the seating strategy require?
Choice and variety of seating

What share of occupied spaces must comply?
75%, plus all classrooms

Whose five principles are used, and what is one?
Kellert and Calabrese

Can Core and Shell projects pursue the thermal or sound environment options?
No

Interior atria can supply up to what share of the required view area?
30%

How many options are available to New Construction projects?
5

How close must occupants be to qualifying glazing?
Within 3 times the head height

What glare and color rendering limits apply?
Luminance under 6,000 cd/m² between 45 and 90 degrees from nadir, CRI 90 or higher

Name some acoustic criteria categories.
Internal and external background noise, masking sound, outdoor environment, reverberation, sound and vibration insulation, impact noise

What share of regularly occupied area earns 2 and 3 points?
75% / 90%

What is the maximum unified glare rating?
19 or lower

How far from the glazing must viewed nature or objects be?
At least 25 ft

Above what ASE must a space get a glare response, and does it leave the sDA average?
Above 20%, and no

What scale classifies acoustic privacy?
High security to no privacy

What space must every project include, and what cannot count?
An accessible quiet space, not a restroom

Beyond ASHRAE 55, what condition must the investigation address?
Overcooling in nontemperate seasons

Can views through bronze-tinted glass count?
Yes, if VLT is above 40% and the view is preserved

Across how many categories are acoustic expectations mapped?
5

How must the biophilic strategies function?
As one integrated composition NOT IN RS

EQC3 · CREDIT

Accessibility and Inclusion

NC 1 PTS | CS 1 PTS | QUALITY OF LIFE

Make the building usable by more people and foster a sense of belonging.

REQUIREMENTS

REQ

Comply With Local Accessibility Codes

Codes identified in IPp2. Without a code, ramps, 32 in clear doors and accessible counters.

AND

REQ

Ten inclusion strategies

At least 10 beyond code across physical diversity, safety and aging, social health, and navigation.

NUMBERS TO KNOW

10	Accessibility strategies beyond code

FROM THE PRACTICE TESTS

12 QUESTIONS

Signage must cover languages spoken by more than what share of the local population?
5%

What does the all-gender restroom strategy require?
One on each floor

What does the door operator strategy require?
Wave or hand and foot press operators

What must emergency alerts provide?
Audible and visual alerts

What do floor-level changes need?
Visual, tactile, contrasting or photoluminescent warnings

What passing spaces do paths narrower than 60 in need?
60 in passing spaces at least every 200 ft NOT IN RS

With no accessibility code, what three basics are required?
Ramps, 32 in doors, accessible counters

If fewer than 10 listed strategies are relevant, what can the team do?
Submit up to 3 alternatives NOT IN RS

Which strategy category holds nonslip flooring?
Safety and aging

Meeting spaces must fit mobility devices for what share of occupants?
10%

Besides accessible equipment, what does the fitness strategy require?
An open, accessible route to and around the equipment

What is required where a stair is an inaccessible route?
An alternate accessible route with the same start and end points

EQC4 · CREDIT

Resilient Spaces

NC 2 PTS | CS 2 PTS | QUALITY OF LIFE

Help occupants stay safe through smoke events, disease and power outages.

REQUIREMENTS

OPT 1 · 1 PTS · NC ONLY

Episodic Outdoor Conditions Mode

ASHRAE Guideline 44 mode for events like wildfire smoke, verified in commissioning.

AND/OR

OPT 2 · 1 PTS · NC ONLY

Respiratory Disease Mode

Infection risk mode meeting ASHRAE 241-2023 equivalent clean airflow.

AND/OR

OPT 3 · 1–2 PTS

Thermal Safety During Power Outages

Designate thermal safety zones that stay habitable during an outage.

PATH 1 · 1 PTS

Consider Extreme Heat

AND/OR

PATH 2 · 1 PTS

Consider Extreme Cold

AND/OR

OPT 4 · 1–2 PTS

Operable Windows

Operable windows in 50% / 75% of occupied spaces, sized per 62.1-2022 section 6.4.

NUMBERS TO KNOW

50% / 75%	Occupied spaces with operable windows

STANDARDS

ASHRAE 62.1-2022
Ventilation and IAQ in commercial spaces

ASHRAE 241-2023
Control of infectious aerosols

ASHRAE Guideline 44
Management mode for wildfire smoke and other outdoor events

TRANSFERS

Resilience line
IPp1 → SSc4 → EAc6 → EQc4

FROM THE PRACTICE TESTS

8 QUESTIONS

Which options can Core and Shell projects use?
Options 3 and 4

What does the credit encourage about hazard timing?
Plan for simultaneous events NOT IN RS

Must the chosen options match the top hazards from IPp1?
No, any two can be chosen, though the assessment should inform the choice NOT IN RS

Is thermal habitability the same as the ASHRAE 55 comfort zone?
No NOT IN RS

Which criteria must the windows meet?
ASHRAE 62.1-2022 Section 6.4

What occupant density limit applies to thermal safety zones?
1 person per 20 sf NOT IN RS

What standard governs the episodic outdoor event management mode?
ASHRAE Guideline 44

How long is the modeled power outage?
2 days at peak summer or winter conditions

EQC5 · CREDIT · NC ONLY

Air Quality Testing and Monitoring

NC 2 PTS | CS NONE | QUALITY OF LIFE

Test and monitor indoor air to support health-based operations.

REQUIREMENTS

OPT 1 · 1–2 PTS

Preoccupancy Air Testing

PATH 1 · 1 PTS

PM and Inorganic Gases

4-hour test for CO, ozone, PM2.5 and PM10 against the limits.

AND/OR

PATH 2 · 1 PTS

Volatile Organic Compounds

ISO 17025 lab tests for the listed VOCs. Formaldehyde 20 µg/m³.

AND/OR

OPT 2 · 1 PTS

Continuous Indoor Air Monitoring

Monitors for CO2, PM2.5, TVOC, temperature and RH, 3 to 6 ft above the floor.

STANDARDS

CDPH Standard Method v1.2-2017
VOC emissions testing of products

ISO/IEC 17025
Lab accreditation

ISO 16000, EPA TO-11a, TO-15, TO-17, ASTM D5197
Indoor air sampling methods

ISO 14644-1
Particle cleanliness classes (Class 8 or lower)

FROM THE PRACTICE TESTS

5 QUESTIONS

Which contaminant uses peak concentration instead of average?
Carbon monoxide

At what height are monitors mounted?
3 to 6 ft above the floor

When may retail projects run preoccupancy air testing?
Within 14 days of occupancy

Which parameters must the monitors measure?
CO2, PM2.5, TVOC, temp, RH

How many measurements does a 120,000 sf office need?
7

PR

Project Priorities

10 points in New Construction · **10** in Core and Shell · the line to Town Hall

☆	PRc1	Project Priorities	9 / 9
🎗	PRc2	LEED AP	1 / 1

Points shown as NC / CS. A dash means the credit is not offered in that system.

PRC1 · CREDIT

Project Priorities

NC 9 PTS | CS 9 PTS | DECARBONIZATION | QUALITY OF LIFE | ECOLOGICAL CONSERVATION AND RESTORATION

Reward regional priorities, project-type needs, exemplary performance, pilots and innovation.

REQUIREMENTS

REQ · 1–9 PTS

Any combination

Mix the sources below for a maximum of 9 points.

Regional Priority
Credits USGBC flags as important for the project's region.

AND/OR

Project-Type Credits
Credits for the needs of a specific adaptation or building application.

AND/OR

Exemplary Performance
Going well past an existing v5 prerequisite or credit.

AND/OR

Pilot Credits
Credits from USGBC's pilot library.

AND/OR

Innovation Strategies
Measurable performance from a strategy LEED v5 does not address. State intent, requirements and submittals.

NUMBERS TO KNOW

9	Max points for regional, project-type, exemplary, pilot and innovation credits

FROM THE PRACTICE TESTS

7 QUESTIONS

What four items must the team identify?
Intent, requirements, submittals, approach

How does exemplary performance differ from a project-type credit?
Exceeds a credit / unique building needs

Regional priority credits address which kinds of priorities?
Environmental, social equity, public health

What must the team document?
The regional priority, with evidence and the strategies used
NOT IN RS

Where does a regional priority credit come from?
USGBC's Credit Library

What do project-type credits address?
Unique needs for the given adaptation or building application

What referenced standards apply?
None NOT IN RS

PRC2 · CREDIT

LEED AP

NC 1 PTS | CS 1 PTS

Encourage the team integration a LEED AP brings and streamline certification.

REQUIREMENTS

REQ · 1 PTS

LEED AP

At least one principal participant holds a LEED AP with a specialty that fits the project.

NUMBERS TO KNOW

1	Point for a LEED AP on the team

FROM THE PRACTICE TESTS

1 QUESTIONS

When must the LEED AP hold an active credential?
At the time of certification review NOT IN RS